

**Basic Principles of Medicine 2**  
**Module: Digestion and**  
**Metabolism      DLA**

**Histology of the Gastrointestinal**  
**System – Overview**

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# Digestive System

Consist of:

1. The **alimentary canal** which **begins @ mouth** and **ends @ anal canal**. **Contains** the **GI tract** which **transport ingested food**, **modifies** it for easy to **absorbs @ different sites**
2. **Associated** organs such as **digestive glands, tongue** and **teeth**

**FUNCTIONS of the alimentary canal are therefore:**

- **Physical & Chemical digestion**
- **Absorption**
- **Excretion** of digestive waste
- **Secretion** of **hormones, digestive juices** and **electrolytes**
- **Physical barrier** – **prevents entry** of **noxious** substances, **antigens** and **pathological organisms**
- **Immunological protection**

**ORIGIN of the alimentary canal:**

- The **epithelium** of the alimentary canal and its **associated glands** is derived from the **ENDODERM** of the **primitive gut tube**
- This is **true for all areas of the canal EXCEPT epithelia of the oral cavity** and **anal canal** which are derived from **ECTODERM**
- The **CT & Muscular tissue** components are derived from **MESODERM**.
- The **peritoneal linings** which covers some of the organs is specifically from the **lateral plate mesoderm**

# Alimentary Canal – Gastrointestinal tract

The **GI tract extends from Proximal Esophagus to Anal canal**. It has the same basic structure throughout its length. The tract is organized in **four distinct layers**. Below these layers are listed based on their **proximity to the lumen** of the canal

1. Mucosa
2. Submucosa
3. Muscularis Externa
4. Serosa / Adventitia

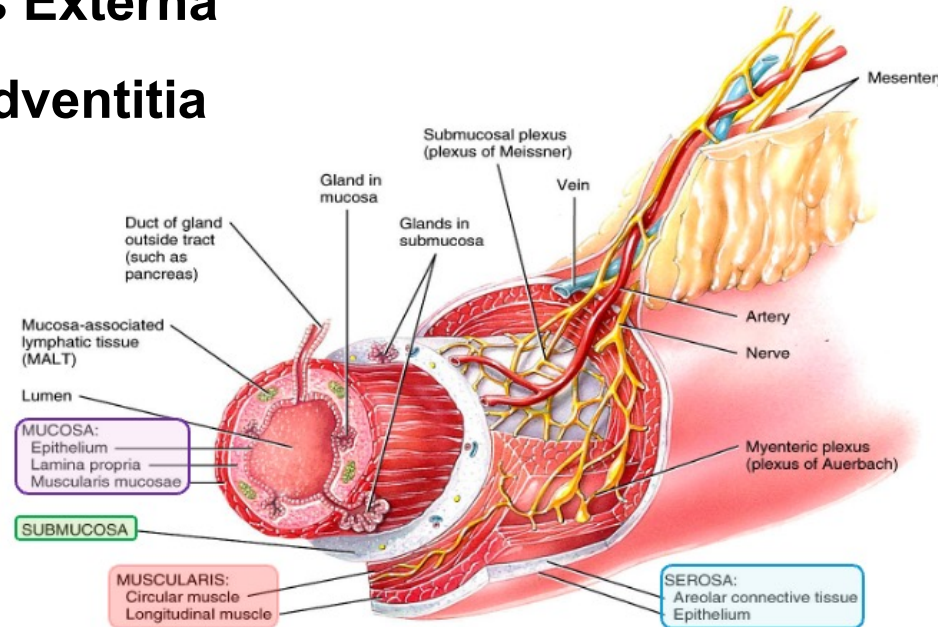
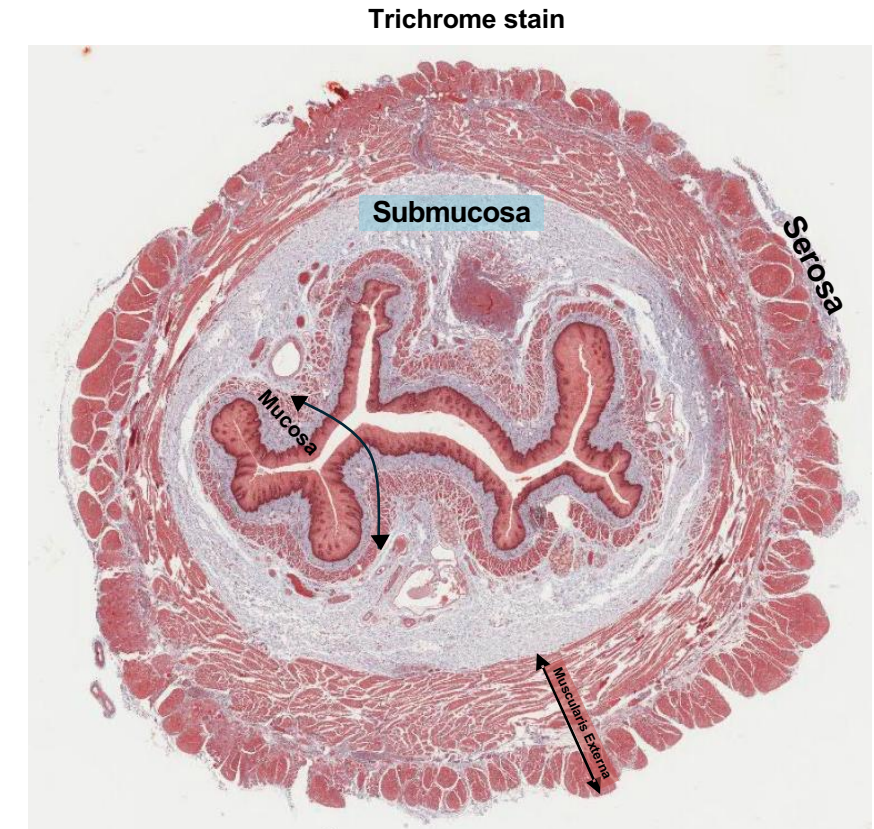


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# Mucosa

The mucosa varies considerably from region to region however three distinct layers is present:

1. **Epithelium**
2. **Lamina Propria** contains **LCT + Blood vessels + Lymph vessels + mucosal Glands + GALT**
3. **Muscularis Mucosae** forms the **boundary between mucosa & submucosa**. It is usually **formed by 2 layers of Smooth Muscle, outer longitudinal & inner circular (however, there is an exception)**. **Contraction produces movement of the mucosal components and changes the surface area for absorption or secretion**

The mucosa is in immediate contact with the food in the lumen. Its three main functions are:

- **Absorption** = of digested **nutrients, water & electrolytes** that enter **into blood and lymph vessels**
- **Secretion** = of **mucus, digestive enzymes, hormones & antibodies**, facilitated by **various glands**
- **Protection** = **Physical barrier by separating the lumen (continuous with the external environment) from tissues and organs of the body (internal environment)**, also serves **immunological protection by lymphatic tissue** serves the body's **first line of immune defense**

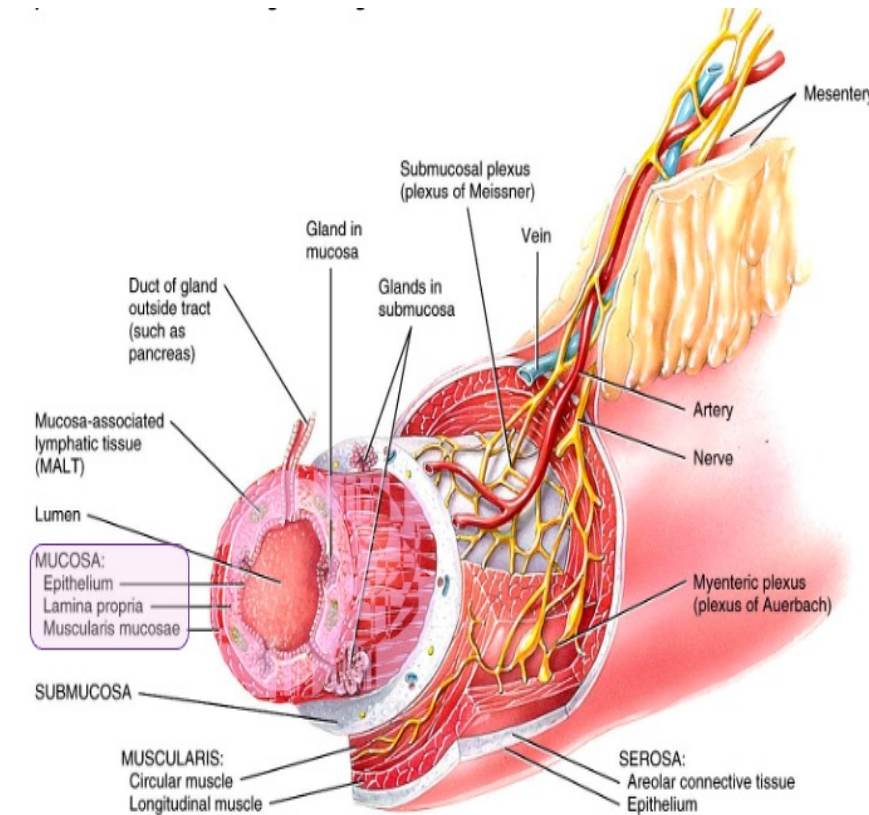


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# Submucosa

- The Submucosa **LIES** just **deep** to the **Mucosa**
- The Submucosa lies **BETWEEN** the **Muscularis Mucosae** & **Muscularis Externa**.

## Typical components includes

- **Dense irregular CT**
- **Large blood vessels & lymphatic vessels**  
= site of esophageal varices & anal varices
- **Some** areas contain submucosal **glands**
- **Contains a part of the Enteric NS =**  
**Meissner's Submucosal Plexus**

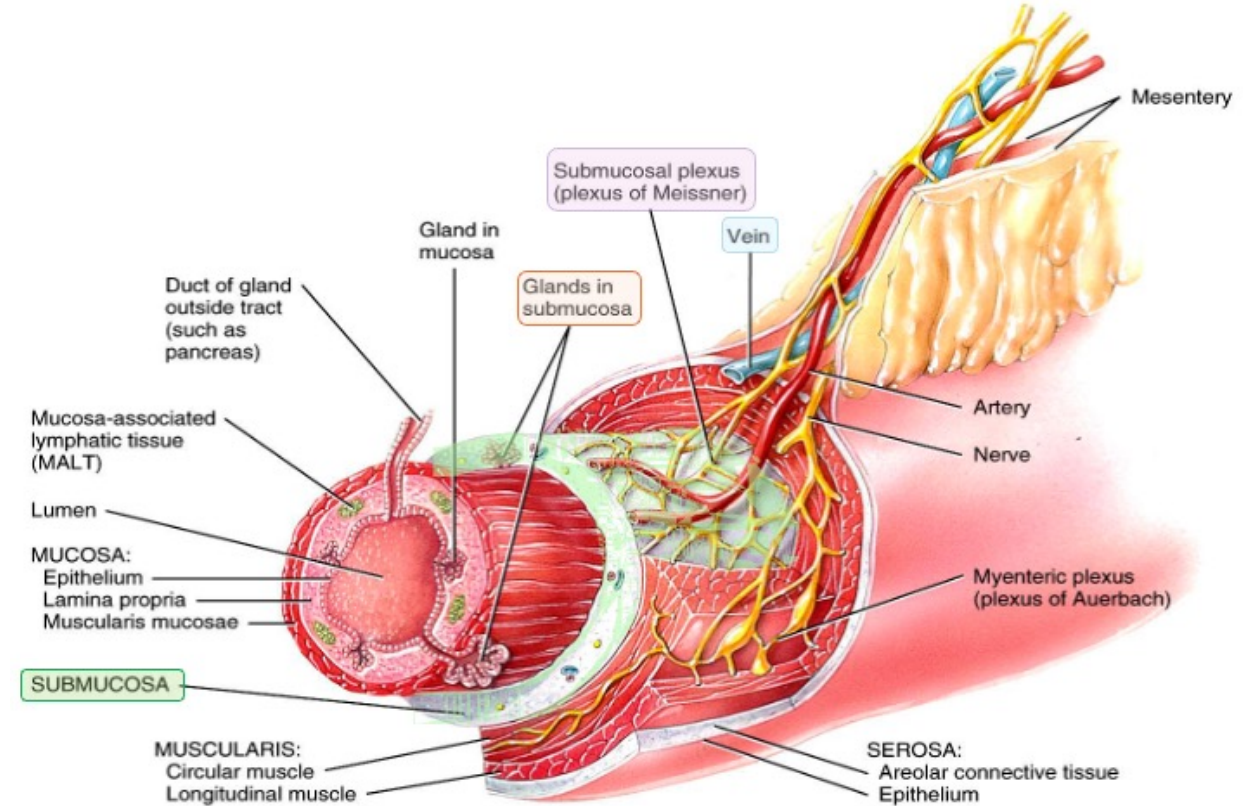
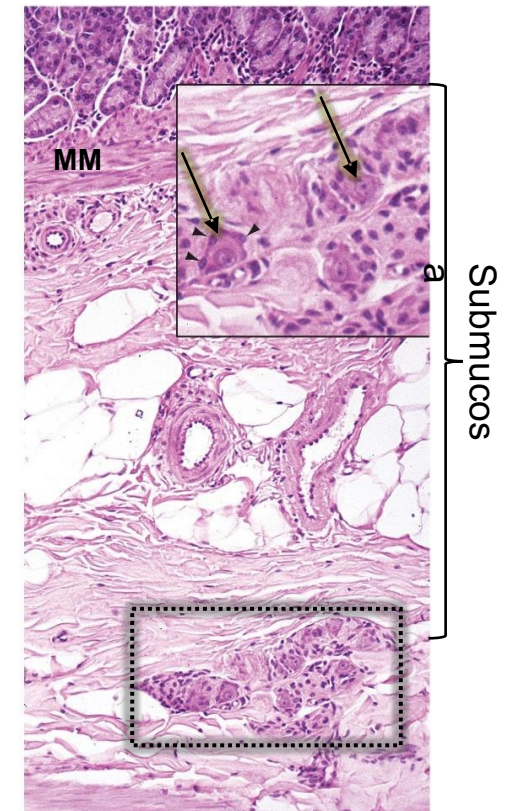
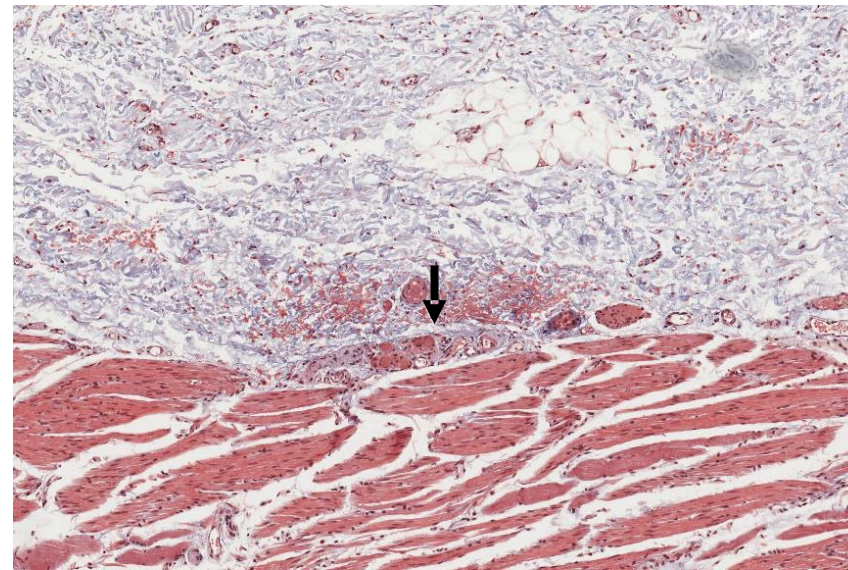
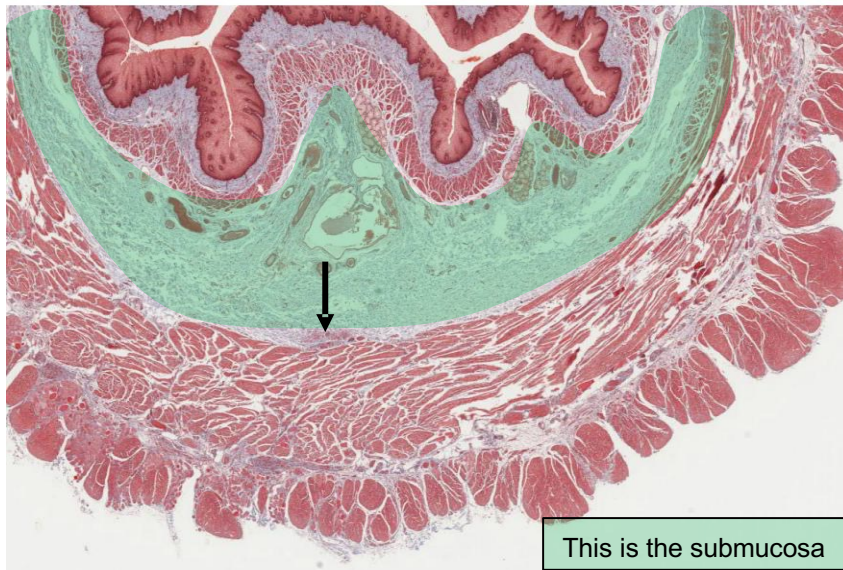


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# Meissner's Submucosal Plexus

- **Meissner's Submucosal Plexus** = can be identified as **CLUSTERS** of neuronal cell bodies located in **submucosa** (arrows)
- **Cell bodies** = **POST-ganglionic PARAsympathetic** neurons and innervate **Muscularis MUCOSAE (MM)** and other mucosal components



Task: What do you think is the embryological precursor to this structure?



# Muscularis Externa

- The Muscularis Externa **LIES** just **deep** to the **Submucosa**
- The Muscularis Externa **or simply muscularis** typically **contains two thick layers of Smooth Muscle**

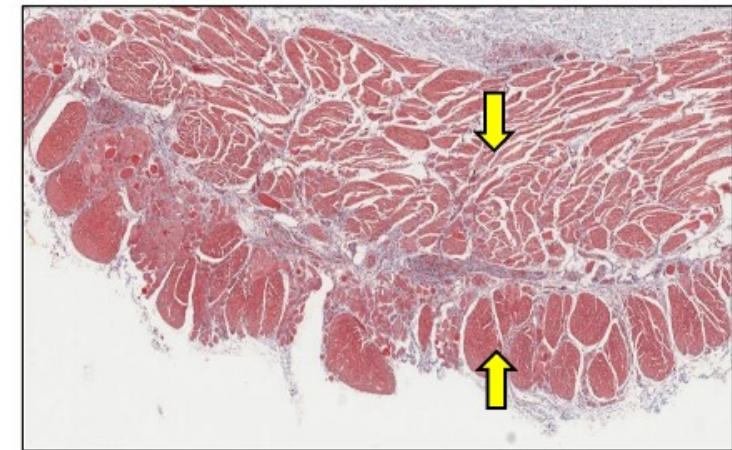
## Inner Circular Layer

- **CONTRACTION = COMPRESSES** and **MIXES** content by **constricting** the lumen
- **Forms SPHINCTERS** at specific locations
  - **Superior Esophageal Sphincter (Pharyngo-esophageal sphincter)**
  - **Inferior Esophageal Sphincter**
  - **Pyloric sphincter**
  - **Ileocecal valve**
  - **Internal Anal Sphincter**
- **These sphincters** may be either the **result of thickening of the circular layer** or **due to an increase** in muscular **tonicity**

## Outer Longitudinal

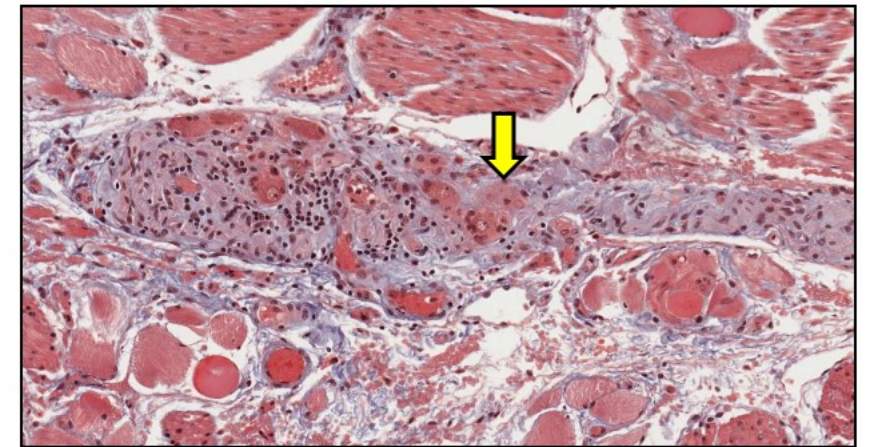
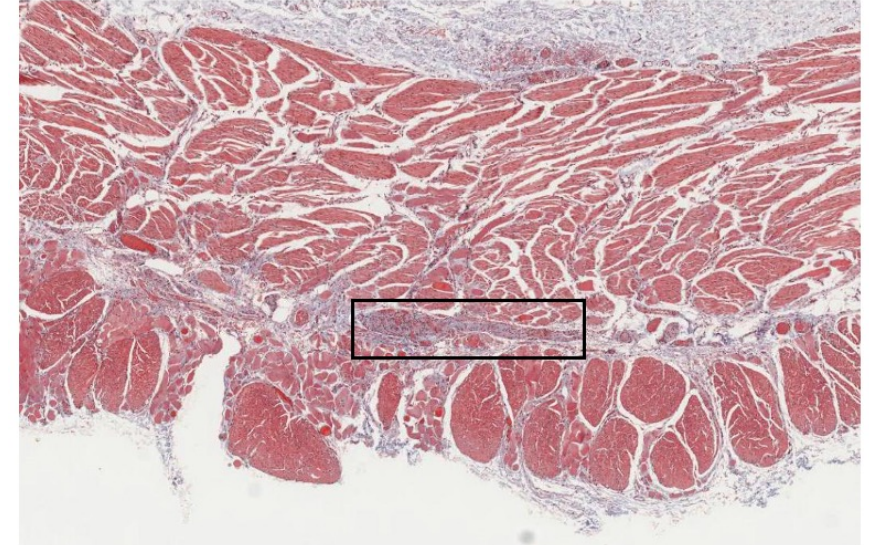
- **Contractions = PROPELS** content of lumen **forward** by **shortening** the **tube**

**Contraction of these** muscle layers **produces peristalsis** (waves of contraction) which **moves content along** the **GI tract**



# Myenteric Auerbach's Plexus

- The **Myenteric Auerbach's Plexus** also forms **part of the Enteric NS**
- It is always **located between** the **circular** muscle layer and **longitudinal** muscle layers of the **Muscularis Externa**
- Contains **POST-ganglionic PARAsympathetic** neurons
- It **innervates** the **Muscularis Externa** for **SLOW, RHYTHMIC** contraction (**peristalsis**)



**Pale staining Parasympathetic ganglion cells located between inner circular (CM) and outer longitudinal (LM) muscle layers**



# Adventitia & Serosa

Figure A: This slide is a cross section through the GI Tract

- **Serosa = Serous Membrane = CT** that is **lined by simple squamous epithelium / mesothelium**
- **Mesothelium produces** a **small amount of fluid** which **allows friction free sliding of abdominal organs** over each other once the serosa is intact, **injury** to this can **result in adhesions**
- **Organs with a complete serous lining** (referred to as **visceral peritoneum** in the **GI**) are referred to as **intraperitoneal** and are **suspended by a mesentery** (double layer of peritoneum)
- **Large blood vessels, nerves and lymphatics** travel **through the mesentery & serosa** to reach the **inner layers of the wall of the digestive tract**



Figure B: This slide is a cross section through the Superior Mediastinum

- **Adventitia is the External CT** that **covers** of an **organ or blood vessels**. It **protects & attaches structures to the body wall**.
- **Parts of the GI tract that do not possess a complete serosal covering = called retroperitoneal**
- **Observe the CT fascia connecting the esophagus & trachea**

